Darrin Bright

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Research Interests

Medical Image Analysis | 3D Vision | Multimodal Learning

Education

Vellore Institute of Technology (VIT) Vellore, India
Integrated Master of Technology in Software Engineering *2022 – 2027 (Expected)*

CGPA: 8.81/10.0
Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Data Mining, Database Management Systems, Python Programming

Publications

[1] **D. Bright**, R. Raj, K. Keisham. "PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation." *Accepted at Conference on Vision and Intelligent Systems (CVIS)*, 2025. [Paper][Code]

Research Experience

Indian Institute of Technology, Kharagpur Kharagpur, India
Research Intern | Advisor: Dr. Debanjan Das *May 2026 – Present*

- Developing GraphUNet, a graph-augmented U-Net for cuffless blood pressure estimation, integrating beat-level morphological graph representations with multichannel ECG and PPG waveform modelling.

Indian Institute of Technology, Hyderabad Hyderabad, India
Research Intern | Advisor: Dr. C. Krishna Mohan *January 2026 – Present*

- Developing a federated continual learning framework for medical image classification in which clients generate privacy-preserving synthetic samples for replay, mitigating catastrophic forgetting under non-IID, privacy-constrained aggregation.

Simon Fraser University Vancouver, Canada
Research Intern | Advisor: Dr. Ghassan Hamarneh *December 2025 – Present*

- Developing a deep learning pipeline for automated detection of mitochondria–endoplasmic reticulum contact sites in STED microscopy volumes.
- Generating training data from both real STED acquisitions and a synthetic pipeline that renders parametric organelle masks into realistic STED-like images via the pySTED simulator.

Vellore Institute of Technology Vellore, India
Undergraduate Researcher *July 2025 – June 2026*

SPHINX-Net: A Lightweight Histology-Aware Framework for OSCC Detection

Advisor: Dr. Bharanidharan N

- Developed SPHINX-Net, a lightweight hybrid framework with histology-aware multi-view contrastive pretraining, enabling strong patch-level OSCC detection.
- Designed a spectral-aware recalibration block that reweights channels by frequency-domain energy to target the nuclear-texture irregularity central to tumour grading.
- Introduced a prototype-graph head that resolves ambiguous patches through mini-batch graph context while enforcing compact class geometry, without whole-slide-image graphs.

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation

Advisor: Dr. Kanchan Keisham

- Developed PortionNet, a cross-modal knowledge distillation framework enabling RGB models to learn 3D geometric features from point clouds during training, eliminating depth sensor requirements at inference.

- Designed a dual-mode training strategy with a lightweight RGB-to-Geometry Adapter that learns to generate pseudo-3D features for accurate geometric reasoning from a standard RGB image.
- Achieved state-of-the-art performance on MetaFood3D and SimpleFood45 datasets, demonstrating effective generalization.

Indian Institute of Technology, Jodhpur Jodhpur, India
Research Intern | Advisor: Dr. Deepak Mishra *May 2025 – July 2025*

Self-Supervised Medical Image Segmentation with Vector Quantization

- Developed a hierarchical vector quantization framework that addresses limited labeled data in medical imaging by learning discrete representations from unlabeled ISIC 2018 images and using them as structured regularizers during segmentation fine-tuning on the PH2 dataset.
- Designed three-stage training pipeline integrating VQ-VAE, SimCLR contrastive learning and segmentation fine-tuning, with multi-layer quantization capturing features across semantic scales.
- Achieved 79.76% Dice, a +1.38% consistent improvement over baseline without VQ across multiple runs.

Ablation Study: Vector Quantization Integration in Transformer-based Segmentation

- Conducted a 13-experiment ablation study on Pascal VOC 2012 dataset investigating vector quantization integration with Swin Transformer encoders, testing codebook configurations, fusion mechanisms and initialization strategies.
- Achieved 74.61% Dice score with optimized vector quantization configuration, demonstrating +7.34% improvement over baseline CNN and +3.91% over vanilla Swin Transformer, validating VQ effectiveness for semantic segmentation.

Technical Skills

Programming: Python, SQL
ML/DL Frameworks: PyTorch, TensorFlow, MONAI
Computer Vision: OpenCV, PointNet, Albumentations, Torchvision
Libraries & Tools: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn

Honors & Awards

- **Runner-up**, Startup Demo Day, Vellore Institute of Technology, 2025
- **Second Runner-up**, HackWar Hackathon, Vellore Institute of Technology, 2025
- **Winner**, Project 2039 Hackathon, Vellore Institute of Technology, 2024
- **Winner**, Alphaforge Ideathon, Vellore Institute of Technology, 2024
- **Runner-up**, Biomimicry Innovation Challenge, Vellore Institute of Technology, 2024

Service & Leadership

IEEE Robotics and Automation Society, VIT Vellore, India
Chairperson *December 2024 – December 2025*

- Led a 500+ member student chapter, organizing technical workshops, hackathons and research talks, while mentoring junior students through hands-on sessions on ML fundamentals and Generative AI frameworks (LLMs, LangChain).

National Service Scheme Vellore, India
Volunteer *January 2023 – January 2024*

- Conducted career guidance sessions for government school students on educational pathways and professional opportunities.